LECTURE 1

Foundations: Stochastic Processes and ARMA Models

Prof. Pedro Valls Pereira pedro.valls@fgv.br

FGV EAESP · MSc / PhD in Business Administration

Agenda

01 Stochastic processes & stationarity

02 Autocovariance & autocorrelation

03 White noise

04 Moving average processes MA(q)

05 Autoregressive processes AR(p) & PACF

06 ARMA / ARIMA processes

07 Box–Jenkins: identification, estimation, validation

08 Forecasting & model evaluation

Throughout the lecture we use the daily USD/BRL exchange rate (Jan 2000 – Mar 2003) as the running empirical example — from raw series through model selection to out-of-sample forecasts.

Section 1

Stochastic Processes & Stationarity

A **stochastic process** is a random function indexed by time $t \in T$ and outcome $\omega \in \Omega$:

$$X : T \times \Omega \longrightarrow \mathbb{R}, \qquad (t, \omega) \mapsto X(t, \omega).$$

Parametric example

$$X(t, \omega) = Y(\omega) \cos\big[Z(\omega) + t \cdot U(\omega)\big]$$

with $Y \sim N(0, \sigma_y^2)$, $Z \sim N(0, \sigma_z^2)$, $U \sim U[-\pi, \pi]$ mutually independent.

For each t we observe **only one realisation** x_t. We cannot recover the distribution of X_t without further restrictions.

Stochastic Processes II

Two restrictions needed:

- **Temporal homogeneity** — statistical structure stable over time *(stationarity)*
- **Limited memory** — distant past gives negligible information about the present *(ergodicity)*

Two Notions of Stationarity I

Strict Stationarity

For any $\{t_1, \ldots, t_n\} \subset T$ and any $\tau \in \mathbb{Z}$:

$$D(X_{t_1}, \ldots, X_{t_n}) = D(X_{t_1+\tau}, \ldots, X_{t_n+\tau})$$

The *entire joint distribution* is invariant to time shifts.

Strong condition — rarely tested directly. Implies weak stationarity when second moments exist.

Two Notions of Stationarity II

Weak (Covariance) Stationarity

Three conditions:

$$(1)\ \mathbb{E}(X_t) = \mu < \infty$$
$$(2)\ \mathrm{Var}(X_t) = \gamma(0) < \infty$$
$$(3)\ \mathbb{E}[(X_{t_1} - \mu)(X_{t_2} - \mu)] = \gamma(t_2 - t_1)$$

Covariance depends *only on the lag*, not on t.

The workhorse assumption throughout this course.

A stationary process looks **statistically identical through any time window**. Slide the window forward or backward — the joint distribution you see never changes.

Section 2

Autocovariance & Autocorrelation Functions

Autocovariance & Autocorrelation Functions I

Autocovariance function

$$\gamma(s) = \mathbb{E}\big[(X_t - \mu)(X_{t-s} - \mu)\big]$$

Properties:

- $\gamma(0) \geq 0$
- $|\gamma(s)| \leq \gamma(0)$ (Cauchy–Schwarz)
- $\gamma(s) = \gamma(-s)$ (even function)
- $\gamma(\cdot)$ is non-negative definite

Autocorrelation function (correlogram)

$$\rho(s) = \frac{\gamma(s)}{\gamma(0)} \in [-1, 1]$$

Autocovariance & Autocorrelation Functions II

Autocovariance generating function
Assuming $\sum_s |\gamma(s)| < \infty$:

$$g_X(z) = \sum_{j=-\infty}^{+\infty} \gamma(j)\, z^j$$

Section 3

White Noise

White Noise I

$\{X_t\}$ is **white noise** if:

$$\mathbb{E}(X_t) = \mu, \quad \mathrm{Var}(X_t) = \sigma^2, \quad \gamma(s) = \begin{cases} \sigma^2 & s = 0 \\ 0 & s \neq 0 \end{cases}$$

Equivalently ($\mu = 0$): $\mathbb{E}(X_t \mid \mathrm{I}_{t-1}) = 0$
$\Rightarrow$ *completely unpredictable from its own past.*

White Noise II

Testing for white noise

Individual lags (homoskedastic):

$$\hat{\rho}_s \overset{a}{\sim} N\left(0, \tfrac{1}{T}\right), \quad \text{95\% band: } \pm \frac{1.96}{\sqrt{T}}$$

Ljung–Box joint test:

$$Q^* = T(T+2)\sum_{k=1}^{m} \frac{\hat{\rho}_k^2}{T-k} \overset{a}{\sim} \chi^2_m$$

under $H_0 : \rho(1) = \cdots = \rho(m) = 0$

White noise is the **benchmark for residual tests** throughout the course.

Section 4

Moving Average Processes

Moving Average Processes MA(q) I

$$X_t = \mu + u_t + \theta_1 u_{t-1} + \cdots + \theta_q u_{t-q}, \qquad u_t \overset{\text{iid}}{\sim} (0, \sigma^2)$$

Moments

$\mathbb{E}(X_t) = \mu$

$\gamma(0) = (1 + \theta_1^2 + \cdots + \theta_q^2)\sigma^2$

$\gamma(s) \neq 0$ for $s = 1, \ldots, q$

$\gamma(s) = 0$ for $s > q$

ACF signature

ACF cuts off sharply after lag q.

PACF decays geometrically.

Spike count in ACF $= q$.

Moving Average Processes MA(q) II

Invertibility

Roots of $\theta(z) = 0$ must lie **outside** the unit circle.

Then $MA(q)$ has an $AR(\infty)$ representation.

Moving Average Processes MA(q) III

$X_t = u_t + \theta u_{t-1}$:

$$\rho(1) = \frac{\theta}{1+\theta^2}, \qquad \rho(s) = 0 \text{ for } s \geq 2$$

For $\theta = -0.5$: $\rho(1) \approx -0.4$

$X_t = u_t + \theta_1 u_{t-1} + \theta_2 u_{t-2}$:

$$\rho(1) = \frac{\theta_1 + \theta_1\theta_2}{1+\theta_1^2+\theta_2^2}, \quad \rho(2) = \frac{\theta_2}{1+\theta_1^2+\theta_2^2}$$

$\theta_1 = -0.5, \theta_2 = 0.25$: $\rho(1) \approx -0.476$, $\rho(2) \approx 0.190$

Section 5

Autoregressive Processes & PACF

Autoregressive Processes AR(p) I

$$X_t = \mu + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + u_t \qquad \Leftrightarrow \qquad \phi(L)\, X_t = \mu + u_t$$

Stationarity condition

All roots of $\phi(L) = 0$ must lie **outside** the unit circle.
Guarantees $MA(\infty)$ representation via Wold's theorem.

Wold's decomposition:

$$X_t = \psi(L)\, u_t, \qquad \psi(L) = \frac{1}{\phi(L)}$$

Autoregressive Processes AR(p) II

AR(1) key results ($|\phi| < 1$)

$$\mathbb{E}(X_t) = \mu/(1-\phi)$$
$$\gamma(0) = \sigma^2/(1-\phi^2)$$
$$\rho(s) = \phi^s \quad \text{(exponential decay)}$$

Autoregressive Processes AR(p) III

Yule–Walker equations give $\rho(s)$ for AR(p):

$$\rho(s) = \phi_1 \rho(s-1) + \cdots + \phi_p \rho(s-p), \quad s \geq 1$$

ACF signature

ACF decays exponentially (or oscillates).
PACF cuts off after lag p.
Spike count in PACF $= p$.

Partial Autocorrelation Function I

ϕ_{kk} is the correlation between X_t and X_{t-k} **after removing the linear effects of all intermediate lags**. It equals the last coefficient in OLS of X_t on $X_{t-1}, \ldots, X_{t-k}$.

Model	ACF	PACF	Identification
AR(p)	Decays exponentially	Cuts off after lag p	PACF spikes $= p$
MA(q)	Cuts off after lag q	Decays exponentially	ACF spikes $= q$
ARMA(p, q)	Decays after lag q	Decays after lag p	Both tails decay
ARIMA(p, d, q)	Slow / non-zero	Unit-root pattern	Difference first

At lag 1: $\phi_{11} = \rho(1)$

At lag 2: $\phi_{22} = \dfrac{\rho(2) - \rho(1)^2}{1 - \rho(1)^2}$

For higher lags: Durbin-Levinson recursion.

Partial Autocorrelation Function II

Stationarity: roots of $\phi(z) = 0$ outside unit circle.
Invertibility: roots of $\theta(z) = 0$ outside unit circle.

Section 6

ARMA & ARIMA Processes

ARMA(p, q) and ARIMA(p, d, q) Processes I

$$\phi(L)\, X_t = \mu + \theta(L)\, u_t, \qquad \phi(L) = 1 - \phi_1 L - \cdots - \phi_p L^p, \qquad \theta(L) = 1 + \theta_1 L + \cdots + \theta_q L^q$$

Key properties

- $\mathbb{E}(X_t) = \mu/(1 - \phi_1 - \cdots - \phi_p)$
- For lags $> q$: ACF behaves like pure AR(p)
- For lags $> p$: PACF behaves like pure MA(q)

ARMA(p, q) and ARIMA(p, d, q) Processes II

ARIMA(p, d, q): ARMA(p, q) applied to the d-th difference of X_t. An integrated process has a characteristic root on the unit circle.

If $\phi \approx \theta$ in ARMA(1,1): polynomials share near-common roots $\Rightarrow$ ACF/PACF $\approx$ white noise $\Rightarrow$ model collapses to ARMA(0,0). **Always inspect roots after estimation.**

Section 7

Box–Jenkins: Identification, Estimation & Validation

The Box–Jenkins Approach I

Step 1 Identification

- Determine order d (unit root tests)
- Inspect ACF / PACF plots
- Apply parsimony principle
- Use IC over full ARMA grid

Step 2 Estimation

- AR: OLS (conditional MLE) or exact MLE
- MA / ARMA: iterative backcasting
- Exact MLE: unconditional dist. of first p obs.
- Cond. MLE: treats first p obs. as fixed

The Box–Jenkins Approach II

Step 3 Validation

- Residuals: white noise? Normality?
- CUSUM: break in mean
- CUSUMSQ: break in variance
- Recursive parameter estimates

If diagnostics **fail** $\Rightarrow$ return to Step 1. The cycle repeats until residuals pass all tests.

Estimation: MLE vs. Conditional MLE I

Exact MLE — AR(1)

Uses unconditional distribution of X_1:

$$X_1 \sim N\left(0,\ \frac{\sigma^2}{1-\phi^2}\right)$$

Log-likelihood:

$$\begin{aligned}\log l = &-\tfrac{T}{2}\ln(2\pi) - \tfrac{T}{2}\ln\sigma^2 \\ &- \tfrac{1}{2\sigma^2}\sum_{t=2}^{T}(X_t - \phi X_{t-1})^2 \\ &+ \tfrac{1}{2}\ln(1-\phi^2) - \tfrac{(1-\phi^2)}{2\sigma^2}X_1^2\end{aligned}$$

Non-linear $\Rightarrow$ Newton–Raphson:

$$\theta_{i+1} = \theta_i - H^{-1}(\theta_i)\, q(\theta_i)$$

Estimation: MLE vs. Conditional MLE II

Conditional MLE (OLS) — AR(1)

Treats first p obs. as fixed. Closed form:

$$\hat{\phi} = \frac{\sum_{t=2}^{T} X_t X_{t-1}}{\sum_{t=2}^{T} X_{t-1}^2}, \qquad \hat{\sigma}^2 = \frac{1}{T}\sum_{t=2}^{T}(X_t - \hat{\phi} X_{t-1})^2$$

No closed form. Initialise $\hat{u}_0 = 0$, build $\hat{u}_t(\theta)$ recursively for given θ, then minimise $\sum \hat{u}_t^2$.

Differences between exact and conditional MLE are **negligible for large** T.

Information Criteria for Model Selection I

Let $k = p + q + 1$ and $\hat{\sigma}^2$ be the MLE variance estimate.

AIC (Akaike 1974)

$$\ln \hat{\sigma}^2 + \frac{2k}{T}$$

Not consistent. Tends to select larger models.

BIC (Schwarz 1978)

$$\ln \hat{\sigma}^2 + \frac{k \ln T}{T}$$

Strongly consistent. Heavier penalty. **Preferred in practice.**

Information Criteria for Model Selection II

HQIC (Hannan-Quinn 1979)

$$\ln \hat{\sigma}^2 + \frac{2k \ln \ln T}{T}$$

Consistent. Intermediate penalty.

7×7 grid of ARMA(p, q) with $p, q \leq 7$. Minimum BIC at $(p, q) = (0, 2)$: $-6.2108 \Rightarrow$ **AR(2) selected**. Two complex roots imply a pseudo-periodic pattern. Keep estimation sample constant when comparing.

Validation: Recursive Residuals & Structural Stability I

OLS residuals are heteroskedastic and serially correlated **by construction** — they cannot be used to test white noise. Use **recursive (standardised) residuals** instead.

One-step forecast error (recursive residual):

$$v_{t+1} = y_{t+1} - \hat{\beta}_t' x_{t+1}, \qquad \mathrm{Var}(v_{t+1}) = \sigma^2 f_{t+1}, \qquad \tilde{v}_{t+1} = \frac{v_{t+1}}{\sqrt{f_{t+1}}} \overset{\text{iid}}{\sim} N(0, \sigma^2)$$

CUSUM

$W_t = \sum_{j=k+1}^{t} \tilde{v}_j / s$

Detects breaks in the **conditional mean**.
$\mathbb{E}(W_t) = 0$ under constancy.

Validation: Recursive Residuals & Structural Stability II

CUSUMSQ

$$S_t = \frac{\sum_j \tilde{v}_j^2}{\sum_{\text{all}} \tilde{v}_j^2}$$

Detects breaks in the **variance**.
Expected path: straight line.

Recursive estimates

Plot $\hat{\beta}_t \pm 2$ s.e. over time.

If initial CI covers final estimate $\Rightarrow$ parameters constant.

Section 8

Forecasting & Model Evaluation

Forecasting from ARMA Models I

Optimal l-step forecast: $X_{T+l|T} = \mathbb{E}[X_{T+l} \mid I_T]$

AR(1)

$X_{T+l|T} = \phi^l X_T$

MSE $= \sigma^2(1 + \phi^2 + \cdots + \phi^{2(l-1)})$

Reverts to mean as $l \to \infty$.

Wide PI only if $|\phi| \to 1$.

MA(1)

$X_{T+1|T} = -\theta_1 \hat{u}_T$

$X_{T+l|T} = 0$ for $l \geq 2$

Collapses to mean in **one step**.

MSE : σ^2 then $(1 + \theta_1^2)\sigma^2$.

Forecasting from ARMA Models II

ARIMA(0,1,0) + drift

$X_{T+l|T} = X_T + l\mu$

$\mathrm{Var}(\upsilon_{T+l}) = l\sigma^2$

Uncertainty **grows linearly** with horizon.

Wide PI always.

General MSE (via MA(∞) representation):

$$\mathrm{MSE}(X_{T+l|T}) = \sigma^2(1 + \psi_1^2 + \cdots + \psi_{l-1}^2)$$

When model fitted on $Y_t = \ln X_t$: $\hat{X}_{T+l|T} = \exp\big(\hat{Y}_{T+l|T} + \frac{1}{2}\mathrm{Var}(\upsilon_{T+l})\big)$

Forecast Evaluation Metrics I

Metric	Formula	Interpretation
RMSFE	$\sqrt{\sum (X_{t\|T} - X_t)^2 / h}$	Scale-dependent; compare same variable
MAE	$\sum \lvert X_{t\|T} - X_t \rvert / h$	Less sensitive to outliers
MAPE	$\sum \lvert X_{t\|T} - X_t \rvert / \lvert X_t \rvert / h$	Scale-free; undefined if $X_t = 0$
Theil U	$\frac{\text{RMSFE}}{\sqrt{\sum \hat{X}^2/h} + \sqrt{\sum X^2/h}} \in [0,1]$	0 = perfect fit

MSE decomposition:

$$\text{MSE} = \underbrace{(\bar{\hat{X}} - \bar{X})^2}_{\text{bias proportion}} + \underbrace{(s_{\hat{X}} - s_X)^2}_{\text{variance proportion}} + \underbrace{2(1 - \rho_{\hat{X},X}) s_{\hat{X}} s_X}_{\text{covariance proportion}}$$

Applied Example

USD/BRL Exchange Rate

Applied Example: USD/BRL Exchange Rate I

Dataset: `bolsa.xls` — Daily Jan 2000 to Mar 2003 (Bovespa, IBX, BRL/USD, S&P 500)

Steps 1 & 2 — Identification

- All series in levels: ACF decays slowly $\Rightarrow$ stochastic trend
- ADF test on log(Dolar): fails to reject $I(1)$
- $\Delta \log(\text{Dolar})$ is stationary; ACF/PACF shows structure up to lag 7
- BIC over 7×7 grid $\Rightarrow$ **AR(2)** (BIC $= -6.2108$)
- Two complex roots $\Rightarrow$ pseudo-periodic dynamics

Steps 3 & 4 — Validation & Forecasting

- CUSUM: no break in conditional mean
- CUSUMSQ: **variance instability** at obs. ≈ 300 and ≈ 650
- Normality: **rejected**
- Forecast: last 7 obs. withheld; all fall within 95% PI but bands are wide

Applied Example: USD/BRL Exchange Rate II

Residuals show non-constant variance and non-normality. This motivates **ARCH/GARCH** modelling.

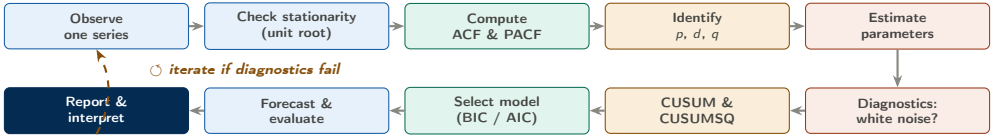
BIC Grid for $\Delta \log(\text{Dolar})$ I

AR\MA	0	1	2	3	4	5	6	7
0	−6.1699	−6.1924	**−6.2108**	−6.2032	−6.1997	−6.1928	−6.1854	−6.1853
1	−6.2043	−6.2017	−6.2029	−6.1961	−6.1932	−6.1883	−6.1805	−6.1773
2	−6.2059	−6.1992	−6.1992	−6.1936	−6.1851	−6.1785	−6.1760	−6.1691
3	−6.2022	−6.1996	−6.1938	−6.1855	−6.1782	−6.1837	−6.1765	−6.1609
4	−6.1998	−6.1920	−6.1857	−6.1811	−6.1830	−6.1766	−6.1687	−6.1714
5	−6.1918	−6.1866	−6.1895	−6.1837	−6.1765	−6.1680	−6.1700	−6.1639
6	−6.1839	−6.1794	−6.1835	−6.1764	−6.1682	−6.1745	−6.1635	−6.1624
7	−6.1869	−6.1787	−6.1706	−6.1625	−6.1759	−6.1728	−6.1633	−6.1550

Rows = AR order (p), columns = MA order (q). **Highlighted: global minimum AR(2).**
Note: an ARMA(0,2) achieves the minimum BIC, but AR(2) has the same value and is the most parsimonious formulation after checking for common roots.

Keep the **estimation sample constant** when comparing models across the grid. Varying the sample changes T and makes BIC values incomparable.

Lecture 1 — Full Workflow Summary I



Key concepts

Stationarity: time-stable moments
ACF/PACF: identify p and q
White noise: residual benchmark

Model signatures

MA(q): ACF cuts off at q
AR(p): PACF cuts off at p
ARMA: both tails decay

Lecture 1 — Full Workflow Summary II

Watch out for

Near-cancellation of roots
Non-constant variance
$\Rightarrow$ GARCH (Lecture 5)

Problem Set — Exercises I

1. For the IBV series in `bolsa.xls`: determine the best ARIMA order, estimate it, and validate residuals using CUSUM and CUSUMSQ.
2. Repeat Exercise 1 for the IBX and S&P 500 series.
3. **Monte Carlo:** Compare conditional MLE and exact MLE for AR(1) with $\phi = -0.9,\ 0,\ +0.9$. How do they differ for small T?
4. **Monte Carlo:** Repeat Exercise 3 for MA(1) with $\theta = -0.9,\ 0,\ +0.9$.
5. Show that the first two partial autocorrelations of AR(2) are estimators of ϕ_1 and ϕ_2.
6. Given the ACF/PACF table below (T=1000), identify the ARIMA class and suggest parameter estimates:

Lag	1	2	3	4	5	6	7	8	9	10
ACF	0.825	0.587	0.388	0.241	0.151	0.110	0.084	0.057	0.039	0.048
PACF	0.825	-0.295	0.016	-0.018	0.035	0.036	-0.028	-0.017	0.017	0.072

7. For the IPCA series (`ipca.xls`): determine the integration order, estimate the best ARIMA, and test stability with recursive OLS.

Core References I

Core textbooks

- Hamilton, J.D. (1994). *Time Series Analysis*. Princeton UP.
- Cochrane, J.H. (2005). *Time Series for Macro & Finance*. Chicago. *[free online]*
- Box, G.E.P. et al. (2015). *Time Series Analysis*, 5th ed. Wiley.
- Tsay, R.S. (2010). *Analysis of Financial Time Series*, 3rd ed. Wiley.
- Lütkepohl, H. (2005). *New Introduction to Multiple Time Series Analysis*. Springer.
- Morettin, P.A. & Valls Pereira, P.L. (2021). *Econometria Financeira*, 4th ed. Blucher.

Core References II

Applied papers (Lecture 1)

- Liu, Y., Shankar, V. & Yun, W. (2017). Price promotion strategies. *Journal of Marketing*, 81(6), 50–66.
- Graham, R.C. & Frankenberger, K.D. (2011). Earnings effects of marketing communication. *Journal of Advertising*, 40(2), 5–24.
- Hyndman, R.J. & Athanasopoulos, G. (2021). *Forecasting: Principles & Practice*. OTexts. *[free online]*

Software: R, Python or Stata. Problem sets require ACF/PACF, unit root tests, ARIMA estimation and recursive diagnostics.

Next: Lecture 2

Non-Stationarity: Unit Roots and Trends

ADF · Phillips–Perron · KPSS · Trend-stationarity vs. difference-stationarity